STRUCTURAL RESILIENCE AND RECOVERY DYNAMICS OF SOCIAL MEDIA NETWORKS: A CASE STUDY ON THE REDDIT ECOSYSTEM

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This research investigates the structural resilience and recovery dynamics of social media networks, specifically focusing on the Reddit ecosystem. In an era where digital communication platforms are vital, understanding how these networks withstand and recover from targeted disruptions is crucial. This study employs a multi-phase graph-theoretic approach to evaluate the impact of removing central hubs and the subsequent restoration of global connectivity using synthetic bridge nodes (bots). In the first phase, a network of 30,402 nodes and 29,255 edges was constructed using R, revealing a scale-free topology with high modularity (Louvain: 0.9422). A simulated targeted attack, involving the removal of the top 20 high-degree nodes, resulted in a significant 30.7% loss in edge connectivity and fragmented the network from 2,425 to 10,361 isolated components. The second phase utilized MATLAB to implement a restoration strategy by injecting synthetic bridge nodes to reconnect the fragmented components. The results demonstrated that the strategic placement of 100 bots increased the giant component size by approximately 21.4%, effectively expanding the core reachable network from 12,019 to 14,589 nodes. In the final phase, Python-based evaluation metrics confirmed that while global connectivity was restored, the average path length within the largest component slightly increased to 10.87, reflecting the integration of previously distant sub-communities. This research concludes that a minimal number of synthetic nodes can substantially mitigate the effects of network fragmentation, providing a theoretical framework for maintaining communication efficiency in disrupted social networks.

Keywords: Graph Theory, Network Resilience, Reddit Dataset, Social Media Analysis, Synthetic Bridge Nodes, Targeted Attacks.